- Software Life Cycle Processes
  - Agreement Processes
    - Establish and manage agreements between acquirers and suppliers.
    - Acquisition Process
      - Purpose: Obtain products/services per acquirer's requirements.
      - Outcomes
        - Request for supply prepared.
        - Supplier selected; agreement established.
        - Product/service accepted; obligations satisfied.
      - Activities and Tasks
        - Prepare acquisition strategy, request for supply.
        - Advertise, select supplier, negotiate agreement.
        - Monitor execution, accept deliverables.
    - Supply Process
      - Purpose: Provide products/services meeting agreed requirements.
      - Outcomes
        - Acquirer identified; response prepared.
        - Agreement established; product/service provided.
        - Obligations satisfied; responsibility transferred.
      - Activities and Tasks
        - Prepare supply strategy, respond to requests.
        - Negotiate agreement, execute and deliver.
        - Support post-delivery, close agreement.
  - Organizational Project-Enabling Processes
    - Support project execution via organizational infrastructure and policies.
    - Life Cycle Model Management process
      - Purpose: Define and maintain life cycle models and processes.
      - Outcomes
        - Policies/procedures established; responsibilities defined.
        - Models/processes assessed; improvements implemented.
      - Activities and Tasks: Establish process policies, assess execution, implement improvements.
    - Infrastructure Management process
      - Purpose: Provide infrastructure/resources for projects.
      - Outcomes
        - Infrastructure requirements defined and provided.
        - Resources maintained and updated.
      - Activities and Tasks: Identify infrastructure needs, acquire/maintain resources.
    - Portfolio Management process
      - Purpose: Initiate and manage projects aligned with organizational strategy.
      - Outcomes
        - Projects identified, authorized, and resourced.
        - Projects monitored, redirected, or terminated.
      - Activities and Tasks: Define project portfolio, evaluate viability, close projects.
    - Human Resource Management process
      - Purpose: Provide and develop human resources.
      - Outcomes
        - Skills identified; personnel provided and trained.
        - Conflicts resolved.
      - Activities and Tasks: Identify skill needs, develop training, manage personnel assignments.
    - Quality Management process
      - Purpose: Ensure quality objectives are met.
      - Outcomes
        - Quality policies/objectives defined; evaluation criteria established.
        - Results analyzed; improvements implemented.
      - Activities and Tasks: Plan quality activities, evaluate performance, address non-conformities.
    - Knowledge Management process
      - Purpose: Create and reuse organizational knowledge assets.
      - Outcomes
        - Knowledge assets developed/acquired and shared.
        - Usage data gathered; assets maintained.
      - Activities and Tasks: Plan knowledge strategy, capture/share knowledge, manage assets.
  - Technical Processes
    - Define, develop, and maintain system functionality.
    - Business or Mission Analysis process
      - Purpose: Define business problems and solution spaces.
      - Outcomes
        - Problem/opportunity space defined; solutions analyzed.
        - Preferred solution selected; traceability established.
      - Activities and Tasks: Analyze business needs, characterize solutions, evaluate alternatives.
    - Stakeholder Needs and Requirements Definition process
      - Purpose: Identify and define stakeholder requirements.
      - Outcomes
        - Stakeholders identified; needs transformed into requirements.
        - Agreement obtained; traceability established.
      - Activities and Tasks: Identify stakeholders, elicit needs, define requirements, validate with stakeholders.
    - System/Software Requirements Definition process
      - Purpose: Transform stakeholder requirements into technical requirements.
      - Outcomes
        - System/software requirements defined; constraints identified.
        - Requirements analyzed; traceability established.
      - Activities and Tasks: Define functional/non-functional requirements, analyze feasibility, manage requirements.
    - Architecture Definition process
      - Purpose: Develop system architecture alternatives.
      - Outcomes
        - Architecture views/models developed; stakeholders' concerns addressed.
        - Elements and interfaces defined; baseline established.
      - Activities and Tasks: Define architecture strategy, develop models/views, assess alternatives.
    - Design Definition process
      - Purpose: Develop detailed design of system elements.
      - Outcomes
        - Design characteristics defined; requirements allocated.
        - Alternatives assessed; artifacts developed.
      - Activities and Tasks: Develop design strategy, define elements/interfaces, assess alternatives.
    - System Analysis process
      - Purpose: Analyze technical performance and feasibility.
      - Outcomes
        - Analysis needs identified; results provided for decisions.
        - Traceability established.
      - Activities and Tasks: Plan analysis, perform technical analysis, report results.
    - Implementation process
      - Purpose: Realize system elements per design.
      - Outcomes
        - System elements implemented; verified/validated.
        - Records maintained; traceability established.
      - Activities and Tasks: Prepare implementation, develop elements, verify/validate, package/store.
    - Integration process
      - Purpose: Assemble system elements into a functional system.
      - Outcomes
        - Integration strategy defined; elements integrated.
        - Interfaces checked; anomalies resolved.
      - Activities and Tasks: Plan integration, assemble elements, verify interfaces, manage results.
    - Verification process
      - Purpose: Provide evidence of conformance to requirements.
      - Outcomes
        - Verification strategy defined; system verified.
        - Anomalies identified; traceability established.
      - Activities and Tasks: Plan verification, execute tests, analyze results, resolve issues.
    - Transition process
      - Purpose: Prepare system for operational use.
      - Outcomes
        - Site prepared; system installed/activated.
        - Training provided; transition documented.
      - Activities and Tasks: Plan transition, install system, train users, validate operational readiness.
    - Validation process
      - Purpose: Confirm system meets stakeholder needs in operational environment.
      - Outcomes
        - Validation strategy defined; system validated.
        - Evidence provided; stakeholder agreement obtained.
      - Activities and Tasks: Plan validation, execute tests, document results, obtain stakeholder approval.
    - Operation process
      - Purpose: Deliver system services in operational environment.
      - Outcomes
        - Operation strategy defined; services delivered.
        - Performance monitored; support provided.
      - Activities and Tasks: Plan operations, monitor performance, address incidents, support customers.
    - Maintenance process
      - Purpose: Sustain system capability over time.
      - Outcomes
        - Maintenance strategy defined; issues resolved.
        - System updated; records maintained.
      - Activities and Tasks: Plan maintenance, perform repairs/upgrades, manage logistics, monitor performance.
    - Disposal process
      - Purpose: Retire system elements appropriately.
      - Outcomes
        - Disposal strategy defined; elements retired.
        - Environment restored; records archived.
      - Activities and Tasks: Plan disposal, deactivate/remove elements, manage waste, archive records.
  - Technical Management Processes
    - Manage technical aspects of projects.
    - Project Planning process
      - Purpose: Develop and coordinate project plans.
      - Outcomes
        - Project objectives, scope, and schedules defined.
        - Resources committed; plans activated.
      - Activities and Tasks: Define project scope, establish WBS, plan resources/schedules.
    - Project Assessment and Control process
      - Purpose: Monitor project performance and adjust plans.
      - Outcomes
        - Performance measured; deviations analyzed.
        - Corrective actions initiated; project authorized to proceed.
      - Activities and Tasks: Assess progress, analyze variances, initiate corrective actions.
    - Decision Management process
      - Purpose: Support structured decision-making.
      - Outcomes
        - Decisions identified; alternatives evaluated.
        - Preferred solution selected; rationale recorded.
      - Activities and Tasks: Prepare decision strategy, analyze alternatives, make and manage decisions.
    - Risk Management process
      - Purpose: Identify, analyze, and treat risks.
      - Outcomes
        - Risks identified/analyzed; treatment plans implemented.
        - Risks monitored; changes tracked.
      - Activities and Tasks: Plan risk management, identify/assess risks, implement treatments.
    - Configuration Management process
      - Purpose: Manage system configurations and changes.
      - Outcomes
        - Configuration items identified; baselines established.
        - Changes controlled; audits completed.
      - Activities and Tasks: Plan CM strategy, identify items, manage changes/releases.
    - Information Management process
      - Purpose: Manage information throughout the lifecycle.
      - Outcomes
        - Information identified and developed; stored/accessed appropriately.
        - Records maintained; disposal handled.
      - Activities and Tasks: Plan information strategy, develop/maintain records, archive/dispose information.
    - Measurement process
      - Purpose: Collect and analyze data to support decisions.
      - Outcomes
        - Measurement needs defined; data collected/analyzed.
        - Results reported; actions informed.
      - Activities and Tasks: Plan measurements, collect/analyze data, report results.
    - Quality Assurance process
      - Purpose: Ensure process and product quality.
      - Outcomes
        - QA procedures defined; evaluations performed.
        - Incidents resolved; improvements identified.
      - Activities and Tasks: Plan QA activities, evaluate conformance, address issues.